

# Keshav S

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[github.com/KodeWithKeshav](https://github.com/KodeWithKeshav)

## Education

**Amrita Vishwa Vidyapeetham**, Coimbatore — B.Tech in Computer Science and Engineering Jun 2025 – Present

- **CGPA: 9.61 / 10.0** (as of April 2026) | **College CSE Batch Topper**
- **Coursework:** Machine Learning, IoT Systems

**Velammal Bodhi Campus**, Kolapakkam — Secondary and Senior Secondary Education 2016 – 2023

- **10th Grade (2020–2021): 98.2%** | **12th Grade (2022–2023): 94.8%**
- Secured merit scholarships for outstanding academic performance

## Experience

**Software Developer Intern**, BCE Global Tech 2026 – Present

- Contributing to internal software projects with a focus on backend development and system design
- Developing RESTful APIs using **Spring Boot** and **Node.js/Express**, ensuring scalable and maintainable architecture
- Working with databases to design schemas, perform CRUD operations, and optimize data handling
- Building and integrating frontend components with backend services for seamless user experience
- Collaborating in an agile environment to test, debug, and deploy robust software solutions

**IoT Engineer**, Intel IoT Club, Amrita Vishwa Vidyapeetham 2025 – Present

- Developed IoT prototypes using **ESP32**, **STM32**, and **Raspberry Pi** with MicroPython and Wi-Fi automation
- Collaborated on innovation challenges and campus tech workshops focused on IoT applications

**Research Contributor**, Amrita Vishwa Vidyapeetham 2023 – Present

- Co-authored research papers in AI, Smart Systems, and IoT; submitted to national-level journals

**NSS Volunteer**, Amrita Vishwa Vidyapeetham — Activity Link 2023 – Present

- Engaged in social service initiatives, environmental sustainability, and digital literacy programs

## Publications

**SecureAI-Cyber: An AI-Powered Cybersecurity Solution for Scalable Threat Management** [Springer Link] Feb 2025

Ramya G.R (Prof), **Keshav S**, Niharika V, Vimal Harihar S.K — *Published, Springer*

**Personalized Nutrition & Lifestyle Recommendations with Multimodal AI for Alzheimer's Patients** Dec 2024

Uma J (Prof), Bagavathi C (Prof), **Keshav S**, Vimal Harihar S.K — *Book Chapter (Accepted)*

**AI's Influence on Gender Representation and Societal Norms in Emerging Technologies** [IEEE Link] Feb 2025

Ramya G.R (Prof), **Keshav S**, Vimal Harihar S K — *Published, IEEE*

## Projects

**SerpentSense** [Live] [GitHub] (*AI, Computer Vision, Wildlife Safety*) 2025

- AI-powered snake species identification system trained on Indian snake species using deep learning
- Classifies venomous vs non-venomous snakes and provides region-specific first-aid guidance

- Offers emergency contact details and medical response instructions through intelligent image recognition
- MediLink** [GitHub] (*Blockchain, Supply Chain, Healthcare*) 2025
- Blockchain-based pharmaceutical supply chain system built on Ethereum
  - Ensures traceability and prevents counterfeit drugs using smart contracts
  - Automates compliance checks and secures transactions on immutable ledger
- Safari Connect** [Live] [GitHub] (*Mobile App Development, Flutter*) 2025
- Mobile application connecting wildlife enthusiasts with safari experiences
  - Features real-time booking, location tracking, and a user-friendly interface
- Passive Acoustic Monitoring for Wildlife** [Demo] [GitHub] (*Machine Learning, Audio Processing*) 2025
- ML system with **92% accuracy** to classify wildlife sounds for biodiversity analysis
  - Tested on Jungle Nightjar and Barn Owl vocalizations
- Lazarus – Multi-Agent AI Platform** [Demo] (*Generative AI, Distributed Systems*) 2026
- Developed a multi-agent AI system (5 agents) to automatically analyze, migrate, and deploy legacy codebases
  - Built a self-healing, serverless pipeline using AWS Lambda and Docker to detect errors, generate fixes, and redeploy at scale
- CardioSync – Edge AI ECG Classification System** [GitHub] (*Embedded AI, IoT*) 2025
- Built an edge AI system to classify ECG signals into 5 categories using a 1D-CNN model deployed on resource-constrained hardware
  - Achieved over 95% classification accuracy on real-world ECG data with optimized on-device inference
  - Designed a lightweight signal processing pipeline including noise filtering, baseline correction, and feature extraction for efficient performance
- Smart Beehive Monitoring System** [GitHub] (*STM32, IoT, Environmental Sensing*) 2025
- Developed a real-time monitoring unit using **STM32** microcontrollers to track hive health
  - Integrated temperature and humidity sensors to detect environmental anomalies and swarm activity

## Technologies

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**Languages:** C, C++, Python, Java, JavaScript, Solidity, Haskell, Embedded C (Register Level), HTML, CSS, SQL, Dart

**Web & AI Technologies:** SpringBoot, Node.js, Express, MongoDB, React, Next.js, Machine Learning (TensorFlow, PyTorch, Scikit-learn), Deep Learning, Computer Vision, CNNs, LLM Applications, RAG (Retrieval-Augmented Generation), FastAPI, Flutter, Firebase

**Data & Tools:** Pandas, NumPy, PostgreSQL, Git, Docker, REST APIs

**IoT & Embedded Systems:** Arduino, ESP32, STM32, Raspberry Pi, Sensors, Blynk, Serial Communication, Embedded C

**Blockchain & Other:** Ethereum, Smart Contracts, Git

## Certifications & Achievements

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- **AWS AI for Bharat Hackathon (2026)**, advanced to the pre-finals round with Lazarus, a cloud-native multi-agent AI automation platform on AWS
- **Smart India Hackathon** — Selected among the top 50 teams at the university level
- **IIT Delhi Hackathon** — Placed among the top 50 teams for technical problem-solving
- **AI For All Ideathon, NIT Tiruchirappalli** — Selected among top 50 out of 650+ teams across India
- **Certifications – Bharathidasan University** — Completed certified programs in Office Automation, Web Development, and Mobile App Development